

Introduction to Software Engineering

Requirements Analysis

The student team is required to complete the Software Requirements Specification (SRS) document for the assigned course project, following the attached template.



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Software Requirements Specification

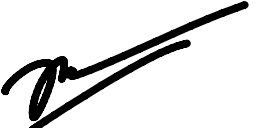
Objectives

This document focus on the following topics:

- ✓ Complete the Software Requirements Specification (SRS) document with the following contents:
 - Elaborate on the Problem Statement
 - Overview of Requirements (Functional and Non-Functional), Stakeholders
 - Use Case Model
 - Use Case Specifications
 - Create Prototype and Mockup Diagrams of the System Interface
- ✓ Đọc hiểu tài liệu phân tích yêu cầu.

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Member Contribution Assessment

ID	Name	Contribution (%)	Signature
23127150	Lê Quốc Anh	100%	
23127205	Lâm Hữu Khánh	100%	
23127215	Trần Tuấn Kiệt	100%	
23127326	Lê Mai Hoài Bảo	100%	
23127407	Trần Hữu Lộc	100%	

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Problem Statement

- *Students describe the project here, providing an approximately 1-2 page(s) business description of the software problem, including the operating environment (e.g., Web browser supporting HTML5, server: Apache...) and design & implementation constraints (e.g., Java programming language, Oracle database, documentation standards...).*

Business Description

Most current social media platforms are specialized, focusing on a single content format. This creates a fragmented user experience.

"Sợ Chỉ City" is an online social network designed to bridge this gap, seamlessly integrating short-form video and image sharing. Its objective is to provide a creative, expressive, and interactive digital space. The platform aims to cultivate a vibrant community where users can share moments, express opinions, and connect with others.

Core functionalities include:

- **Content Management:** Users can upload videos or images, add descriptions and hashtags, and set visibility. Users can also edit, delete, or pin their posts.
- **Social Interaction:** Users can react to posts, leave comments, and share content.
- **Connection:** Users can follow other accounts and build friend lists.

The team's vision is a network for sharing meaningful values and ideas, not just entertainment. It is a step toward building an inclusive and creative online community.

Operating Environment

Client-Side:

- **Platform:** Primarily a web environment.
- **Device:** Desktops, laptops, tablets, or smartphones.
- **Browser:** Any modern web browser supporting HTML5.
- **Network:** Stable internet connection.

Server-Side:

- **Application Server:** Backend running on a Uvicorn server.
- **Infrastructure:** Standard server with sufficient CPU, RAM, and storage.
- **Database:** All data stored in Firestore.
- **File Storage:** User-uploaded images and videos stored in Cloud Storage.

Design & Implementation Constraints

Technological Constraints:

- **Frontend:** Must be built using React, Tailwind CSS, and Shadcn.
- **Backend:** Logic and APIs must be developed with FastAPI.
- **Database:** Must use Firestore as the primary database.
- **Cloud Storage:** Must use Firebase Storage for video and image hosting.
- **Authentication:** Must integrate with Google OAuth Service.

Architectural Constraints: Must use a 3-tier architecture with clear separation between Frontend (React), Backend (FastAPI), and Data.

Process & Documentation Standards:

- **Version Control:** All code and documentation must be stored in GitHub.
- **Directory Structure:** Must follow the specified structure, including /src for code and /docs for documentation (subdivided into management, requirements, analysis/design, and test).
- **Project Management Tools:** JIRA is required. Effective use of these tools is a key grading criterion.

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Requirements Overview

3.1 Stakeholders

STT	Stakeholder	Description
1	Supervisor	The person is both the client and quality supervisor. They provide the project requirements, technical constraints, monitor progress, and are the final decision-maker for the project's evaluation.
2	Development Team	The team responsible for the entire development process: from requirement gathering, system design, and programming (Frontend/Backend) to testing and writing technical documentation.
3	End Users	Consist of two main groups as described in the Use Case: • Guest: A non-registered visitor, only allowed to view and search. • Registered User: A full-access user with abilities to create content, interact, and engage in community activities.
4	External Systems	Third-party platforms that the "Sợi Chỉ City" system relies on for operation: • Google (OAuth): Provides authentication services. • Firestore & Firebase Storage: Provide database and file storage services. If these services change their policies or have any

		problem, the system will be directly affected.
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3.2 Requirements

3.2.1. Functional Requirements Specification

i. Access & Account Management Module

- Related use cases: U001 – U006
- Requirements Specification:
 - **FR-01:** The system must allow Guest users to register a new account. The registration process must support integration with Google OAuth Service for identity verification.
 - **FR-02:** The system must allow users to log in and log out. After a successful login, the system must grant access to functions available only to Registered Users.
 - **FR-03:** The system must provide a password recovery function via email if a user forgets their password.
 - **FR-04:** The system allows users to edit their personal profile (Edit Profile), including updating the avatar image, display name, and bio. Text-based data is stored in Firestore, while images are stored in Firebase Storage.
 - **FR-05:** The system allows users to view profiles (View Profile) with specific access rules and display logic:
 - + Access Control: Both Guest and Registered users can view other users' public profiles. However, accessing "My Profile" requires the user to be logged in.
 - + Display Logic: The profile page must display user information (Avatar, Name, Bio) and their list of posts. Pinned posts (if any) must be displayed at the top, followed by the remaining posts in chronological order.
 - + Privacy: If a profile is private, the system must restrict access to the content for unauthorized viewers.

ii. Content Management Module

- Related use cases: U007 – U009
- Requirements Specification:
 - **FR-06:** The system must provide an interface for users to create new posts (Post Content).
 - + The system must allow uploading short videos or images.
 - + These media files must be securely stored on Firebase Storage.
 - + Users must enter a text caption and can attach hashtags to the post.

- **FR-07:** The system must allow users to manage their posts (Manage Posts):
 - + Users may edit the caption of an existing post.
 - + Users may permanently delete a post from the system.
 - + Users may pin posts they prefer to the top of their personal profile.
- **FR-08:** The system allows users to download media (Download Media), such as images or videos, from a post to their personal device.

iii. Social Interaction Module

- Related use cases: U010 - U016
- Requirements Specification:
 - **FR-09:** The system must provide a Feed Page (View All Posts) to display a list of posts relevant to the user's status.

Access Control:

- + Guest: View Public, Trending, or System-recommended posts.
- + Registered User: View posts from friends (Following) and System recommendations.

Features: The feed must support Infinite Scroll and Pull-to-Refresh to update content.

- **FR-10:** The system must allow users to filter the content source on the Feed page (View with Filter). The system must support a dropdown menu with specific categories:
 - + "For You" (Default): Algorithmic recommendations.
 - + "Following": Posts from followed accounts only.
 - + "Saved": Posts the user has bookmarked.
 - + "Liked": Posts the user has previously liked.
- **FR-11:** The system allows users to send comments on a post (Comment on Post) in the form of text, image, or video.
- **FR-12:** The system must allow users to view specific post details (View Post Detail).
 - + The interface must display the full content (Avatar, Name, Full Text, Timestamp) and support a Full-screen Viewer for attached media (images/videos).
 - + The system must display the list of Comments (direct replies) associated with that post below the main content.
- **FR-13:** The system allows users to react to a post (React to Post). The reaction state must be updated in real-time (toggle Interact/Undo Interact).

- **FR-14:** The system must provide a Post Activity Sheet (View Activity) for Registered Users to view detailed engagement metrics.
 - + Metrics: Display total counts of Likes and Reposts.
 - + Interactors List: Display a list of users who interacted with the post (Avatar, Name, Follow status).
 - + Filter: Users can filter this list by interaction type (Tabs: "Likes", "Reposts").
- **FR-15:** The system provides sharing functions (Share Post) with two modes:
 - + Repost: Creates a shared copy of the original post on the user's personal profile.
 - + External Share: Generates a unique URL and copies it to the clipboard for sharing outside the platform.

iv. Search Engine Module

- Related use cases: U017
- Requirements Specification:
 - **FR-16:** The system must provide a search bar (Search) that works across the entire application. The search results must support three filters:
 - + Search by Hashtag: Filters posts containing specific tags (e.g., #SoiChiCity).
 - + Search by User: Searches for the exact account name or display name of another user.
 - + Search by Context: Searches based on keywords appearing in the content of posts.

v. Community Connection Module

- Related use cases: U018 - U021
- Requirements Specification:
 - **FR-17:** The system allows users to follow other accounts (Follow User) to build a friend list and receive updates from them on the Newsfeed.
 - **FR-18:** The system allows users to unfollow an account (Unfollow User) if they no longer receive updates from them.
 - **FR-19:** The system allows users to block accounts to restrict all interactions and content viewing. This action automatically triggers a mutual unfollow without notifying the blocked user.
 - **FR-20:** The system allows users to unblock accounts via a management list. Unblocking restores access but does not automatically restore the "Following" relationship.

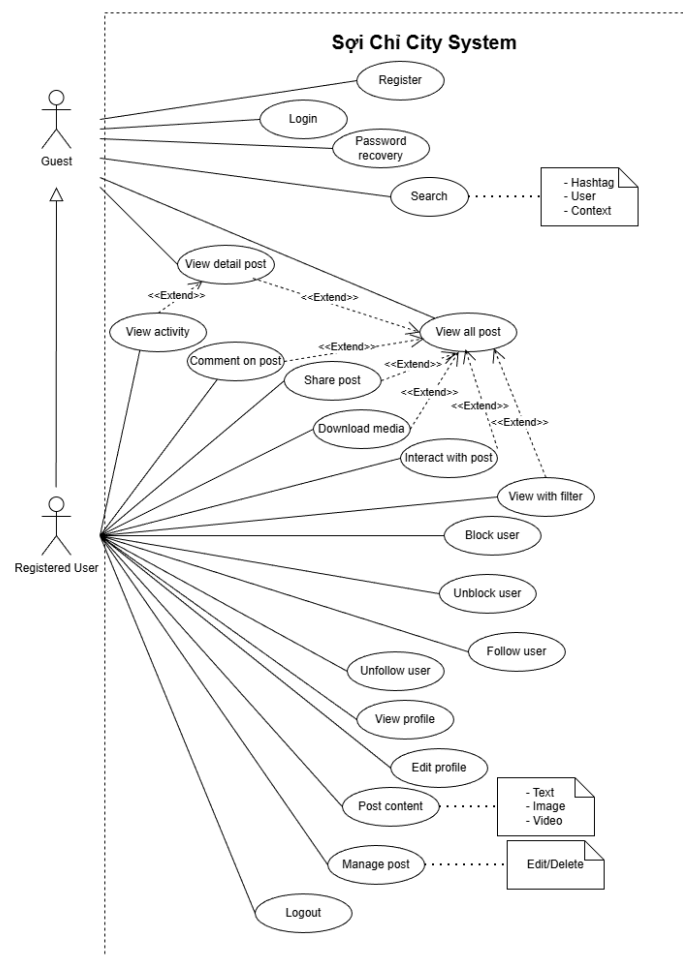
3.2.2. Non-Functional Requirements Specification

- **NFR-01 (Performance):** The backend system (FastAPI + Uvicorn) must ensure an average API response time under 200ms in stable network conditions.
- **NFR-02 (Scalability):** The Firestore database must be designed to scale, capable of handling large amounts of unstructured data as the number of users increases.
- **NFR-03 (Media Storage):** Firebase Storage must be able to handle large amounts of media content, ensuring efficient retrieval and low latency when serving image and video files.
- **NFR-04 (Interface):** The Frontend interface (React + Shadcn) must ensure design consistency and remain user-friendly.
- **NFR-05 (Compatibility):** The web application must be compatible and display properly (Responsive Design) on mobile devices as well as desktop computers.
- **NFR-06 (Maintainability):** The source code must adhere to the defined folder structure (/src, /docs) and follow clean-code principles to allow easy maintenance and future upgrades.

4 Requirements Analysis

4.1 Use Case model

Use Case Diagram of the system (students can draw using software such as StarUML, Visio, Rational Rose, mermaidjs, ...)



4.2 Use Case Specification

4.2.1. Use Case Register

Use case ID	U001
Use Case	Register
Brief Description	Allows new users to create their account to explore more features of the platform.
Actor	New users, Authentication system.
Pre-Condition	1. The user wants to create an account with email that is not already on the system. 2. The Authentication system is online and accessible to display the registration page.
Result	1. A new user account is successfully created and stored in the database. 2. The user can log in using the newly registered credentials and be redirected automatically to the login page.
Main Scenario	1. The user navigates to the registration page. 2. The system displays a registration form (fields include full name, username, email, password, re-enter password). 3. The user fills in all required information. 4. The user clicks the “Sign Up” button. 5. If all data are valid, the system saves the new user’s information into the database. 6. The system displays a “Registration Successful” message and redirects to the Feed page.
Alternative Scenarios	1a. User already has an account 1. The user realizes they already have an account → select “Sign in” instead. 2. The system redirects the user to the Login page. 1b. The user selects “Continue with Google”. 1. The system redirects the user to Google’s authentication page. 2. The user follows the requirements of Google’s authentication page.

	3. The system validates Google's authentication result and redirects the user to the Feed page. 5a. Email already exists 1. The system detects that the email/phone number is already registered. 2. The system shows a message "Email already exists" and asks the user to log in or use a different one. 5b. Invalid password format 1. The system identifies that the password does not meet security requirements (too short, missing special characters, etc.). 2. The system displays an error message "Invalid password format" and requests correction. 5c. Wrong re-enter password. 1. The system identifies that the re-entered password does not match the password. 2. The system displays an error message "Wrong re-enter password" and requests correction.
<i>Non-Functional Constraints</i>	The system must respond with registration results (Success or error) within a maximum of 5 seconds after the user presses the "Sign Up" button under normal network conditions.

4.2.2. Use Case Login

Use case ID	U002
Use Case	Login
Brief Description	Allows registered users to access the platform by providing valid login credentials (e.g., email and password, login with google account).
Actor	Registered user, Authentication system.

<i>Pre-Condition</i>	1. The user already has an account (created manually or via Google). 2. The system is connected to the authentication service (database and Google API are available). 3. The login page is accessible.
<i>Result</i>	1. The user is successfully authenticated and gains access to the platform. 2. The system displays a “Login Successful” message and redirects the user to the Feed page.
<i>Main Scenario</i>	1. The user navigates to the login page. 2. The system displays form login includes email, password and a button “Continue with Google” 3. The user fills information in the form. 4. The user enters their email and password. 5. The user clicks the “Sign In” button. 6. The system validates the input fields (check for empty fields, valid email format, etc.). 7. The system verifies the credentials. 8. If valid, the system authenticates the user and creates a login session. 9. The system redirects the user to the Feed page and displays a success message.
<i>Alternatice Scenarios</i>	3a. The user selects “Continue with Google”. 1. The system redirects the user to Google’s authentication page. 2. The user follows the requirements of Google’s authentication page. 3. The system validates Google’s authentication result and redirects the user to the Feed page. 3b. The user clicks “Forgot Password” The system redirects to the password recovery page and guides the user through resetting their password. 8a. The system detects incorrect email or password. 1. The system displays which field is wrong. 2. The system displays an error message: “Invalid email or password. Please try again.”

Non-Functional Constraints	Authentication and redirection to home page should be completed in under 5 seconds.
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4.2.3. Use Case Logout

Use case ID	U003
Use Case	Logout
Brief Description	Allows a Registered User to securely sign out of the system. This ends their current session to protect their account and personal information.
Actor	Registered User.
Pre-Condition	The user is currently logged into the application.
Result	1. The user's session is securely ended. 2. The user is redirected to the Feed page. 3. The user cannot perform registered user actions anymore until they log in again.
Main Scenario	1. The user clicks the logout button in the sidebar. 2. The system asks for confirmation. 3. The user clicks confirm. 4. The system invalidates the session on the server. 3. The system redirects the user to the Feed page.
Alternative Scenarios	2a. User Cancels 1. The user clicks cancel on the confirmation popup. 2. The system closes the popup. 3. The user remains logged in and stays on the current page.
Non-Functional Constraints	None

4.2.4. Use Case Password Recovery

Use case ID	U004
Use Case	Password recovery
Brief Description	Allows a guest to recover their account by requesting a password reset link via email in case they forget their password.
Actor	Guest user, system
Pre-Condition	At “Guest” mode and already access to the “Forgot password” function.
Result	A password reset link is sent to the user’s email, allowing them to create a new password and regain access to their account.
Main Scenario	1. The user clicks “Forgot Password.” 2. The system displays the password recovery form. 3. The user enters their registered email address. 4. The user clicks “Send reset link.” 5. The system verifies whether the email exists in the database. 6. The system sends a password reset link to the user’s email. 7. The user clicks on the link received in the email. 8. The system redirects them to the password reset page on the website. 9. On this page, the user enters a new password and then re-enters it for confirmation. 10. The system validates the new password and updates it in the database. 11. The user is redirected back to the login screen to log in again.
Alternative Scenarios	3a. User cancels the recovery process 1. The user clicks “Cancel.” 2. The system closes the recovery interface and returns to the login page.

	5a. Email not found or invalid input 1. The system detects that the email does not match any account or is an invalid input. 2. The system displays an error message and requests correction. 6a. System fails to send email (server error, network issue) The system displays: “Unable to send email at the moment. Please try again later.” 9a. Passwords do not match 1. The system detects that the new password and confirmation password are not identical. 2. The system displays: “Passwords do not match” and prompts the user to re-enter. 9b. Password does not meet security requirement (too short, no uppercase, no special character) 1. The system rejects the password based on defined security policy. 2. The system displays a message such as: “Password must meet security requirements.”
Non-Functional Constraints	None

4.2.5. Use Case Edit Profile

Use case ID	U005
Use Case	Edit Profile
Brief Description	Allows a registered user to update their personal

	information (display name, bio, profile picture, cover photo, external links, ...) and modify profile visibility settings (public/private).
Actor	Registered user, system
Pre-Condition	The user must be logged into the system. The user is currently in My Profile Page.
Result	The user's updated profile data and visibility settings are stored in the system database and displayed immediately on their profile page based on the selected visibility mode.
Main Scenario	1. The user clicks "Edit Profile." 2. The system displays the edit profile form. 3. The user updates the desired information. 4. The user selects a profile visibility mode (Public or Private). 5. The user clicks "Save changes." 6. The system validates and saves the new information. 7. The system displays a "Profile updated successfully" notification.
Alternative Scenarios	3a (from step 3). User cancels editing 1. The user clicks the "Cancel" button. 2. The system closes the edit interface without saving any changes. 4a. User does not choose visibility mode The system applies the visibility setting by default (public). 6a. Invalid input (empty name, username is already taken, unsupported image (allowed: PNG, JPEG, JPG, HEIC), etc.) 1. The system detects an error in the input data. 2. The system shows an error message and requests correction.

Non-Functional Constraints	None
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4.2.6. Use Case View profile

Use case ID	U006
Use Case	View profile
Brief Description	Allows any user (guest or registered) to view the profile of other users, depending on the visibility (public or private), from actions such as clicking the username on a post or searching for a user. However, clicking “one’s own profile” button requires authentication (login). Pinned posts, if any, are displayed at the top of the profile.
Actor	User, system
Pre-Condition	To view another user’s profile: no login required. To view your own profile: the user must be logged into the system.
Result	The user successfully views the requested profile (if the visibility of other users permits). If the profile contains pinned posts, they are displayed at the top.
Main Scenario	1. The user navigates to a user’s profile page (via clicking a username, search result, direct link, or “own profile” button). 2. The system displays the user’s public profile information (name, bio, avatar, cover, links, ...). 3. The system displays pinned posts at the top (if any). 4. The system displays the rest of the user’s posts in chronological order.
Alternative Scenarios	1a. User attempts to view their own profile while not logged in 1. The system detects that the user is not authenticated. 2. The system displays a message such as:

	“You must log in to view your own profile.” 3. The system redirects user to the login screen. 2a. User views a private profile 1. The system detects selected portions of the profile are restricted due to privacy settings. 2. The system displays only allowed fields (example: profile picture + name). 3. The system shows a message such as: “This profile is private.” 4a. No pinned posts exist 1. The system skips the pinned section. 2. The system directly displays normal posts only. 5a. No posts exist The system displays a friendly empty-state message, such as: “No posts yet.”
Non-Functional Constraints	None

4.2.7. Use Case Post Content

Use case ID	U007
Use Case	Post (Text, Image, Video)
Brief Description	Allows a registered user to upload a post visible publicly, containing text, image(s), or video(s), optionally with captions and hashtags
Actor	Registered user, system
Pre-Condition	1. The user must be logged into the system. 2. The device must have a stable internet connection.
Result	1. The system stores the post in the database. 2. The post appears on the user’s personal feed page

	and in other users' feed pages.
Main Scenario	1. The user clicks the “Create post” button. 2. The system displays the post creation interface. 3. The user enters text, selects image(s) or video(s) in the attachments tab. 4. The user clicks “Post”. 5. The system validates the uploaded files and content. 6. The system saves the post and displays a “Post successfully created” notification.
Alternative Scenarios	3a (From step 3): User cancels post creation 1. The user clicks the “Cancel” button. 2. The system closes the post creation interface and discards changes. 5a: Invalid file (file type (only accepted: mp4, avi, webm, PNG, JPEG, JPG, HEIC), file size (<= 5mb for image, <= 5 mins for video), ...) 1. The system detects an error in the uploaded file. 2. The system displays an error message and requests the user to reselect valid files.
Non-Functional Constraints	None

4.2.8. Use Case Manage Post

Use case ID	Uoo8
Use Case	Manage Post
Brief Description	Allows the Post Owner to manage their existing posts. The user can perform two main actions: • Edit: Users can update the text content, and remove or replace the attached image/video.

	• Delete: Permanently remove the post from the application. • Pin/Unpin: Pin a specific post to the top of the user's Personal Profile (Timeline) to highlight it, or unpin it to return it to its original chronological position.
Actor	Registered User (Post Owner), System
Pre-Condition	1. The user is logged into the system. 2. The user must be the author of the post they are attempting to manage. 3. The post is currently active and visible.
Result	1. Edit: The post displays the updated text and media content immediately. The system displays an "Edited" indicator (label) on the post. 2. Delete: The post is removed and is no longer visible on any newsfeed or timeline. 3. Pin/Unpin: • If Pinned: The post moves to the very top of the user's Profile/Timeline and displays a "Pinned" icon/label. • If Unpinned: The post returns to its original position based on the creation timestamp.
Main Scenario	SCENARIO A: EDIT POST 1. The user selects the "Options" menu on a specific post. 2. The user selects the "Edit" option. 3. The system displays the post content in an editable mode. 4. The text area is editable. 5. The media section allows options to remove the existing media or replace it with a new image/video. 6. The user modifies the text content and confirms the action (e.g., taps "Save"). 7. The system validates the content. 8. The system saves the changes and updates the post display with the new text and media

	configuration. SCENARIO B: DELETE POST 1. The user selects the "Options" menu on a specific post. 2. The user selects the "Remove" option. 3. The system displays a Confirmation Dialog ("Are you sure you want to delete this post?") to prevent accidental deletion. 4. The user confirms the deletion. 5. The system removes the post from the feed and displays a success notification. SCENARIO C: PIN/UNPIN POST 1. The user selects the "Options" menu on a specific post. 2. The system checks the current status of the post. 3. <i>If the post is currently normal:</i> The option displays "Pin to Profile". 4. <i>If the post is currently pinned:</i> The option displays "Unpin from Profile". 5. The user selects the "Pin" (or "Unpin") option. 6. The system refreshes the User's Profile view: 7. Pin: The post is moved to the top of the list with a visual indicator (Pushpin icon). 8. Unpin: The post acts like a normal post, sorted by date. 9. The system displays a notification ("Post pinned successfully").
Alternative Scenarios	A-5a. Invalid Empty Content (Validation Rule) 1. The user removes all text from a "Text-only" post (a post with no media attached) OR removes both text and media from a media post, and attempts to save. 2. The system validates that the post cannot be empty. 3. The system prevents saving and displays an error message ("Post content cannot be empty").

	A-4a. Cancel Edit: 1. During editing, the user taps "Cancel". 2. The system discards any unsaved changes and returns the user to the previous view. B-4a. Cancel Delete: 1. On the confirmation dialog, the user taps "Cancel". 2. The system closes the dialog; the post remains active. General-1. System/Connection Error 1. The system fails to process the Edit or Delete request due to a system issue. 2. The system displays a friendly error message ("Unable to update post. Please try again later."). 3. The post remains in its original state.
Non-Functional Constraints	None

4.2.9. Use Case Download Media

Use case ID	U009
Use Case	Download Media
Brief Description	Provides functionality for a registered user to save a copy of an image or video from a post to their personal device.
Actor	Registered user, system.
Pre-Condition	1. The user must be logged into the system. 2. The user is viewing a post containing an image/video.
Result	The media file is saved to the local storage on the user's device.
Main Scenario	1. The user clicks on an image or video in the post to view it in fullscreen. 2. The user clicks the icon download in the media viewing interface. 3. The system checks permissions.

	- The system requests the browser to download the file - The browser displays a download complete notification.
Alternative Scenarios	4a User cancels the download. - The system requests the browser to download the file and displays the Save As... dialog. - The user clicks the cancel button in that dialog. - The download action is canceled. 4b Network error during download. - The download is in progress. - The user's Internet connection is interrupted. - The browser automatically pauses or cancels the download displaying Download failed - Network error.
Non-Functional Constraints	Performance: The download process must initiate within 2 seconds after the user confirms permissions.

4.2.10. Use Case View All Posts

Use case ID	U010
Use Case	View All Posts
Brief Description	Allows a user to view posts on the Feed page. Registered User: Views posts from friends and system recommendations. Guest: Views public, trending, or random system-recommended posts. Posts are displayed in chronological order or by relevance.
Actor	Registered user, guest, system

<i>Pre-Condition</i>	1. The device is connected to the internet. 2. There are posts available in the system (Public posts for Guest; Friend/Public posts for User).
<i>Result</i>	The system displays a list of posts relevant to the user's status.
<i>Main Scenario</i>	1. User (Registered/Guest) navigates to the Feed page screen. 2. The system checks the user's authentication status: 3. If User: Retrieves posts from friends and relevant suggestions. 4. If Guest: Retrieves a list of Public or Trending posts only. 5. The system displays the list of posts. 6. Information Display: Author Info, Timestamp, Content (Text/Media), Engagement Status (Count of Likes/Comments). 7. The user scrolls through the list to view the content. 8. Use case ends when the user navigates away or closes the application.
<i>Alternatice Scenarios</i>	2a. System fails to retrieve data (Connection/System Error) 1. System shows a friendly error message and a "Retry" option. 2. User selects "Retry" -> Return to Step 2. 2b. Refresh Feed page (Pull to Refresh) 1. User pulls to refresh. 2. System checks for new data (New friend posts for User; New trending posts for Guest). 3. System updates the list. 4a. View older posts (Load More)

	1. User scrolls to the bottom. 2. System appends the next batch of posts. 4b. Guest attempts to interact (Like/Comment) 1. Guest clicks on "Like", "Comment", or "Share". 2. System detects unauthenticated status. 3. System displays a "Login Required" prompt with options to Login or Cancel. 4. If "Cancel" is selected, the user remains on the Feed page.
Non-Functional Constraints	1. The system must display the list of articles in less than 5 seconds. 2. Use Lazy Loading to load only 10-20 articles per scroll to keep the application running smoothly.

4.2.11. Use Case View Detail Post

Use case ID	U011
Use Case	View Detail Post
Brief Description	Allows the user to view a specific post in detail along with its direct comments. This view focuses solely on the selected post and lists the interaction/replies belonging to it.
Actor	User, system
Pre-Condition	1. The post exists and is accessible (Public or available to the specific user). 2. The device has a network connection. 1. The user is currently viewing the Feed page.
Result	1. The system displays the detailed content of the selected post. 2. The system displays a list of comments associated with that post.

Main Scenario	1. From the Feed page, the User taps on a specific post item (body of the post). 2. The system retrieves the detailed data for the selected post and the list of Comments (direct replies) associated with this post. 3. The System renders the Post Detail View on the screen: • Main Post Section: Displays the full content of the current post (Avatar, Name, Full Text, Media, Timestamp, Stats). • Comments Section: Displays a list of comments below the main post. 4. The User scrolls through the list to read the post content and comments.
Alternative Scenarios	1a. Post is Deleted (Just before click) 1. The User taps on a post in the feed. 2. The system attempts to retrieve details but finds the post has been deleted recently. 3. The system displays a message: "This post is no longer available" and refreshes the feed to remove the item. 5a. View Media 1. The User taps on an attached image or video. 2. The system opens the media in a full-screen viewer. 5b. Interaction Attempt (Login Required) 1. The User attempts to perform an action (Like/Reply). 2. <i>System Logic:</i> If the user is not authenticated (not logged in), the system prompts them to Log In. 3. If authenticated, the action proceeds normally.
Non-Functional Constraints	The system must paginate (paginate) the comment list (for example, only load 10 comments first, the user clicks "See more" to load more) to avoid lag if the article has

	too many comments.
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4.2.12. Use Case View With Filter

Use case ID	U012
Use Case	View With Filter
Brief Description	Allows the user to change the content source displayed on the Feed page via a dropdown menu. The user can switch between the default view for you, following, saved, liked posts.
Actor	Registered User, System
Pre-Condition	1. The user is logged into the system. 2. The user is currently on the Feed page. 3. The filter dropdown is enabled and visible.
Result	1. The system displays the list of posts corresponding to the selected category. 2. The dropdown title updates to reflect the selected filter name.
Main Scenario	1. The user taps the filter dropdown displayed as “For you” by default. 2. The system displays the menu options: “For you” , “Following”, “Saved”, “Liked”. 3. The user selects a specific option. 4. The system closes the menu and validates the selection. 5. The system loads the list of posts matching the selected category. 6. The system refreshes the screen to display the new content and update the dropdown title to the selected filter name.
Alternatice Scenarios	3a. Cancel Filter 1. The user taps outside the dropdown or taps cancel. 2. The system closes the dropdown menu. 3. The process ends; the feed remains unchanged.

	5a. Empty Data 1. The system searches but finds no posts for the selected category. 2. The system displays an empty state message. 3. The system keeps the filter active title shows saved but displays no list items. 5b. Connection Error 1. The system fails to load the data due to network issues. 2. The system displays a generic error message 3. The system keeps the previous list or shows a retry button.
Non-Functional Constraints	Performance: The system must retrieve and display the filtered post list within 3 seconds under normal network conditions. Usability: The dropdown menu must be responsive and clearly highlight the currently active filter option.

4.2.13. Use Case Interact With Post

Use case ID	U013
Use Case	Interact With Post
Brief Description	Allows a registered user to express a "Like" reaction to a post OR "Save" the post to their personal collection. These are toggle actions: 1. Like: Clicking once applies the "Like", clicking again removes it ("Unlike"). 2. Save: Clicking once saves the post ("Save"), clicking again removes it from saved items ("Unsave").
Actor	Registered User, System

Pre-Condition	1. The user must be logged into the system. 2. The post must be visible on the user's screen. 3. The device must have a stable internet connection.
Result	For Like: 1. The system stores the user's "Like" status for this post in the database. 2. The system updates the total "Like" count for the post. 3. The system sends a notification to the post's owner. For Save: 1. The system stores the user's "Save" status for this post in the database (Bookmarked). 2. The post is added to the user's "Saved Posts" list/collection.
Main Scenario	Scenario A: Like Post 1. The user finds a post they wish to "Like" 2. The user clicks the "Like" icon. a. The system immediately updates the UI (Optimistic update): b. Changes the "Like" icon to the liked state (e.g., solid red). 3. Increments the "Like" counter by one temporarily. 4. The system sends a "Like" request to the server. 5. The system validates the request. 6. The system records the "Like" action in the database 7. The system confirms the UI state (ensures the icon remains liked and the counter is accurate). Scenario B: Save Post 1. The user finds a post they wish to "Save" for later viewing. 2. The user clicks the "Save" (Bookmark) icon. 3. The system immediately updates the UI (Optimistic update):

	a. Changes the "Save" icon to the saved state (e.g., solid/filled yellow icon). b. Displays a brief confirmation (e.g., toast message "Post saved"). 4. The system sends a "Save" request to the server. 5. The system validates the request. 6. The system records the "Save" action in the database 7. The system confirms the UI state (ensures the icon remains in the saved state).
Alternatice Scenarios	1a (From Scenario A, Step 2): Unliking a post 1. The user clicks the "Like" icon (which is currently in a liked state). 2. The system immediately updates the UI: reverts icon to unliked state, decrements counter. 3. The system sends an "Unlike" request to the server. 4. The system validates the request. 5. The system removes the "Like" action from the database. 1b (From Scenario B, Step 2): Unsaving a post 1. The user clicks the "Save" icon (which is currently in a saved state). 2. The system immediately updates the UI: a. Reverts the "Save" icon to its unsaved state (e.g., outline only). b. Displays a brief confirmation (e.g., "Post removed from saved"). 3. The system sends an "Unsave" request to the server. 4. The system validates the request. 5. The system removes the "Save" action from the database. 5a (From Step 4 in Scenarios A & B): Connection error 1. The system fails to send the request to the server due to a connection loss. 2. The system displays a temporary error message

	(e.g., "Action failed. Please check your connection."). 3. The UI reverts to its pre-click state (Rollback the optimistic update).
Non-Functional Constraints	None

4.2.14. Use Case Comment Post

Use case ID	U014
Use Case	Comment Post
Brief Description	Allows a registered user to add a comment (reply) to an existing post (thread). The comment can contain text and optionally attach images or videos.
Actor	Registered User, System
Pre-Condition	1. The user must be logged into the system. 2. The original post must be displayed and allow comments. 3. The device must have a stable internet connection.
Result	1. The system stores the new comment in the database, linked to the original post and the user. 2. The new comment appears in the post's comment list. 3. The system sends a notification to the original post's owner (and any @mentioned users).
Main Scenario	1. The user clicks the "Comment" (Reply) icon on a post. 2. The system displays the comment creation interface (text field, attach button). 3. The user enters the comment content (text). 4. The user clicks "Attach" and selects image(s)/video(s). 5. The user presses the enter key on the keyboard or clicks the "Reply" button. 6. The system validates the comment content (e.g., not empty, file attachment validation).

	- The system sends and saves the comment to the database. - The system closes the input interface and updates the comment list to display the new comment. - The system displays a "Comment posted" notification.
Alternatice Scenarios	3a (From Step 3): Cancel comment - The user swipes and clicks outside to dismiss the interface. - The system displays a confirmation prompt "Discard comment?" - The user confirms. - The system closes the comment interface and discards the entered content. 6a (From Step 6): Invalid content - The system detects an error (e.g., empty comment, invalid file type/size). - The system displays a specific error message (e.g., "Comment cannot be empty"). - The user remains on the comment interface to correct the error. 7a (From Step 7): Connection error on send - The system fails to send the comment to the server. - The system displays an error message "Couldn't post a comment. Please try again." - The system saves the comment as a draft to retry sending later.
Non-Functional Constraints	None

4.2.15. Use Share Post

Use case ID	U015
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Use Case	Share Post
Brief Description	Allows a registered user to re-post an existing post onto their personal profile (Repost) or share a link to that post to an external application (External Share).
Actor	Registered user, System.
Pre-Condition	1. The user is logged in. 2. The user is viewing a post. 3. The post allows sharing
Result	1. (If Repost) The system creates a shared copy of the original post on the user's personal page. 2. (If External Share) The post's link is copied to the clipboard.
Main Scenario	1. The user clicks the "Repost" icon under the post. 2. The system checks the user's permissions. 3. The system (after validation) immediately creates a new post (type "repost") on the user's personal page, pointing to the original post. 4. The system displays a "Reposted successfully" notification.
Alternatice Scenarios	1a. External Share Flow ("Copy link") 1. The user clicks the "Send" icon. 2. The system generates a unique URL for the post. The system copies this URL to the device's clipboard. The system displays a "Link copied" notification. 1b. User un repost. 1. The user clicks the "Repost" icon again to un repost the post. 2. The system validates the action then un repost the post for the user. 3. The system displays a "Unreposted successfully" notification.

Non-Functional Constraints	Performance: The system must verify and complete the Repost action within 2 seconds to ensure a smooth user experience
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4.2.16. Use Case View Activity

Use case ID	U017
Use Case	View Activity
Brief Description	Allows the Registered User to view detailed performance metrics of a specific post (Views, Likes, Reposts) and a comprehensive list of accounts that have interacted with that post.
Actor	Registered user, System.
Pre-Condition	1. NoneThe user is logged into the system. 2. The user is currently viewing the post in the View Post Detail.
Result	1. The system displays the Post Activity Sheet overlaying the current screen. 2. The user can visualize the full engagement statistics and the list of interacting users.
Main Scenario	1. From the View Post Detail, User tap “View Activity” below the post content. • The System displays the Activity Sheet appearing over the screen. Activity Sheet will have: ○ Metrics: ▪ Likes: Total number of likes. ▪ Reposts: Total number of reposts/quotes. ○ Interactors: Below the metrics, the System lists the accounts that have interacted with the post (displaying Avatar, Name, and current Follow status). 2. Users can scroll down to see more interactors.

Alternative Scenarios	1a. No Activity Data (Zero State) 1. The User taps to view activity on a newly created post. 2. The System displays all metrics as 0. 3. In the list section, the System displays a placeholder message: "No activity yet" 1b. Data Loading Error (Network Issue) 1. The User taps "View activity". 2. The System attempts to display the sheet but encounters a network error. 3. The System displays a brief error notification (Toast message): "Unable to load activity. Please try again." 4. The User remains on the Post Detail View. 2a. Filter by Interaction Type (Likes/Reposts) 1. On the Activity Sheet, the User taps on a specific tab ("Likes", "Reposts"). 2. The System refreshes the list section to display only the users who performed that specific interaction.
Non-Functional Constraints	None

4.2.17. Use Case Search (User, Hashtag, Context)

Use case ID	U017
Use Case	Search (User, Hashtag, Context)
Brief Description	This function allows users to search for other users, hashtags, or posts by entering keywords in the search bar.
Actor	Registered user, System.
Pre-Condition	None
Result	Search results based on user's input.

Main Scenario	1. The system displays the search bar. 2. The user enters a keyword (username, hashtag, or post content). 3. The system returns matching results. 4. The user selects one result to view details.
Alternative Scenarios	3a. No matching results found 1. The system displays a message: “No matching results found”. 2. The system suggests similar keywords or trending searches.
Non-Functional Constraints	The search interface should support auto-suggestions and be easy to use, followed by these steps: 1. The user starts typing a keyword into the search bar. 2. The system dynamically displays auto-suggestion results based on the entered text. 3. The user can either select one of the suggestions or press Enter / click the Search button to submit the query. 4. The system then returns a full search result page (usernames or posts) based on the final keyword or selected suggestion. Search results must appear within 5 seconds.

4.2.18. Use Case Follow User

Use case ID	U018
Use Case	Follow User
Brief Description	Allow a registered user to "Follow" another user to subscribe to that user's new posts on their own feed.
Actor	Registered user, System.
Pre-Condition	1. The user must be logged in. 2. The user is viewing the profile page of another user.

	3. The user is not currently following the Target User.
Result	1. The database creates a follow relationship (Actor → Target User). 2. The Target User's profile page updates the button to following. 3. The follower count on the Target User's profile increases by 1. 4. The Target User's posts will begin to appear on the Actor's feed.
Main Scenario	1. The user visits the profile page of another user. 2. The system displays the profile information and a follow button. 3. The user taps the follow button. 4. The system receives and validates the request. 5. The system registers the user as a new follower of the target account. 6. The system updates the follower count (increases by 1) on the screen. 7. The system changes the button label from follow to following.
Alternative Scenarios	4a: Connection error 1. The system cannot save the database due to a network error. 2. The system reports an error "An error has occurred, please try again later". 3. The button remains "Follow", and the status has not changed.
Non-Functional Constraints	Performance: The system must validate the request and update the button status within 3 seconds to ensure a smooth user experience. Data Consistency: The follower count must be updated accurately in real-time, even when multiple users follow the target user simultaneously.

4.2.19. Use Case Unfollow User

Use case ID	U019
Use Case	Unfollow User
Brief Description	Allow a registered user to stop following someone they were previously following.
Actor	Registered user.
Pre-Condition	1. The user must be logged in. 2. The user is viewing the profile page of another user. 3. The user is already following this person.
Result	1. The system removes the following connection from the database. 2. The button on the profile page changes back to follow. 3. The follower count on the Target User's profile decreases by 1. 4. The user stops seeing updates from this person in their feed.
Main Scenario	1. The user visits the profile page of the person they want to stop following. 2. The user sees the button labeled "Following". 3. The user clicks the button. 4. The system displays the confirmation board. 5. The system sends a request to remove the relationship. 6. The system processes the request and updates the relationship status. 7. The interface updates immediately, changing the button text back to "Follow".
Alternative Scenarios	3a. Accidental Click / Cancel: 1. The user clicks "Following" but then decides not to unfollow. 2. The user clicks "Cancel" on the popup. 3. Nothing changes. The user is still following the person.

	5a. Connection Error: 1. The user clicks the button, but the internet connection fails. 2. The system shows a small error message. 3. The button stays as "Following".
Non-Functional Constraints	Performance: The system must process the unfollow request and update the UI (button and count) within 2 seconds after the user confirms. Usability (Error Prevention): The system must require a confirmation step (dialog or popup) to prevent users from accidentally unfollowing someone. Data Integrity: The follower count must immediately decrease by exactly 1 to ensure accurate statistics.

4.2.20. Use Case Block User

Use case ID	U020
Use Case	Block User
Brief Description	Allows a registered user to block another user. Once blocked, the blocked user cannot view the blocker's profile, posts or interact with them.
Actor	Registered user, System.
Pre-Condition	1. The user must be logged into the system. 2. The user is viewing the target user's profile or one of their posts. 3. The target user is not currently in the user's block list. 4. The device must have a stable internet connection.
Result	1. The system updates the database: adds the target user's ID to the current user's blacklist. 2. The system automatically executes a mutual unfollow (if applicable). 3. The blocked user cannot find or view the blocker's content. 4. The system displays a notification confirming the block was successful.

Main Scenario	1. The user navigates to the target user's profile (or clicks the options menu on a post owned by the target user). 2. The user clicks the "Options" icon (three dots). 3. The system displays a list of actions. 4. The user selects "Block". 5. The system displays a confirmation dialog with a warning about the consequences (e.g., "They won't be able to message you or find your profile"). 6. The user clicks the "Block" button to confirm. 7. The system processes the request and updates the relationship status in the database. 8. The system hides the blocked user's content from the current view. 9. The system displays a "User blocked" notification.
Alternative Scenarios	6a (From Step 6): Cancel block 1. The user clicks "Cancel" or taps outside the dialog box. 2. The system closes the confirmation dialog. 3. No block action is taken; the user returns to the previous screen. 7a (From Step 7): Connection or System error 1. The system fails to send the block request to the server due to network issues. 2. The system displays an error message: "Could not block user. Please try again later." 3. The relationship status between the users remains unchanged. 7b (From Step 7): Already blocked 1. In rare cases (e.g., action taken on another device), the user is already blocked. 2. The system refreshes the UI state to reflect the blocked status. 3. The system displays a notification: "User is already

	in your block list."
Non-Functional Constraints	None

4.2.21. Use Case Unblock User

Use case ID	U021
Use Case	Unblock User
Brief Description	Allows a registered user to remove a previously blocked user from their blacklist via the Edit Profile section. Once unblocked, the target user can again view the user's profile and interact, but they are not automatically re-followed.
Actor	Registered user, System.
Pre-Condition	1. The user must be logged into the system. 2. The user has at least one account in their block list. 3. The device must have a stable internet connection.
Result	1. The system updates the database: removes the target user's ID from the current user's blacklist. 2. The relationship status returns to "Public" (or non-following). 3. The unblocked user can now view the blocker's content again. 4. The user is removed from the "Blocked Users" list in the interface.
Main Scenario	1. The user navigates to their own Profile. 2. The user clicks the "Edit Profile" button. 3. The system displays the profile editing interface. 4. The user selects the "Block List" menu item. 5. The system displays a list of currently blocked users. 6. The user clicks the "Unblock" button next to the specific user they want to unblock. 7. The system displays a confirmation dialog (e.g.,

	"Are you sure you want to unblock [Name]? They will be able to see your posts."). - The user clicks "Unblock" to confirm. - The system removes the user from the block list in the database. - The system removes the user from the displayed list and shows a "User unblocked" notification.
<i>Alternative Scenarios</i>	4a (From Step 4): Empty Block List - The user clicks "Blocked Users". - The system detects no blocked users. - The system displays an empty state message (e.g., "You haven't blocked anyone yet."). 8a (From Step 8): Cancel Unblock - The user clicks "Cancel" in the confirmation dialog. - The system closes the dialog. - The user remains in the "Blocked Users" list with no changes made. 9a (From Step 9): Connection Error - The system fails to send the unblock request due to network issues. - The system displays an error message: "Failed to unblock. Please check your connection." - The user remains in the list as blocked.
<i>Non-Functional Constraints</i>	None

5

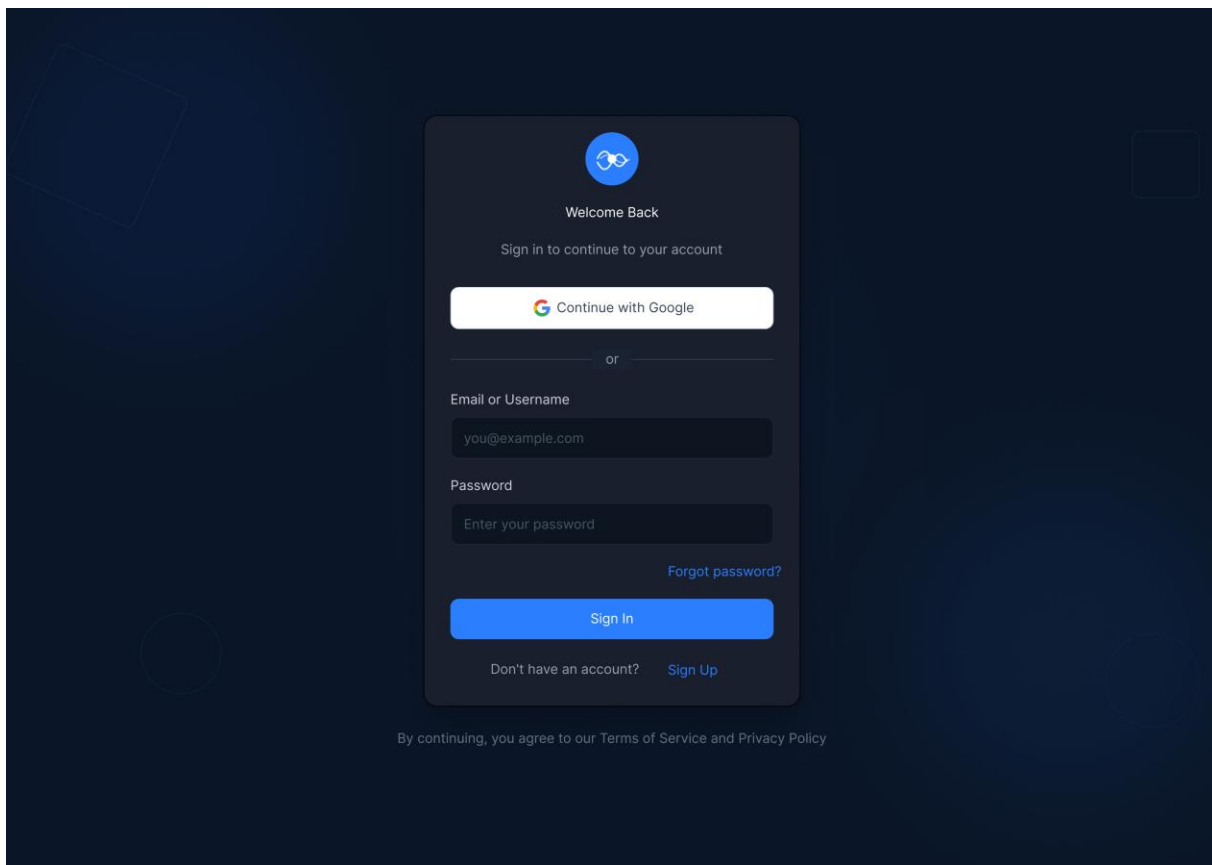
Prototype/Mockup

- The student team should present the software's prototype and wireframe diagrams in this section, which can be created using tools like **Figma**, Pencil, Axure RP, Balsamiq Mockups, etc.

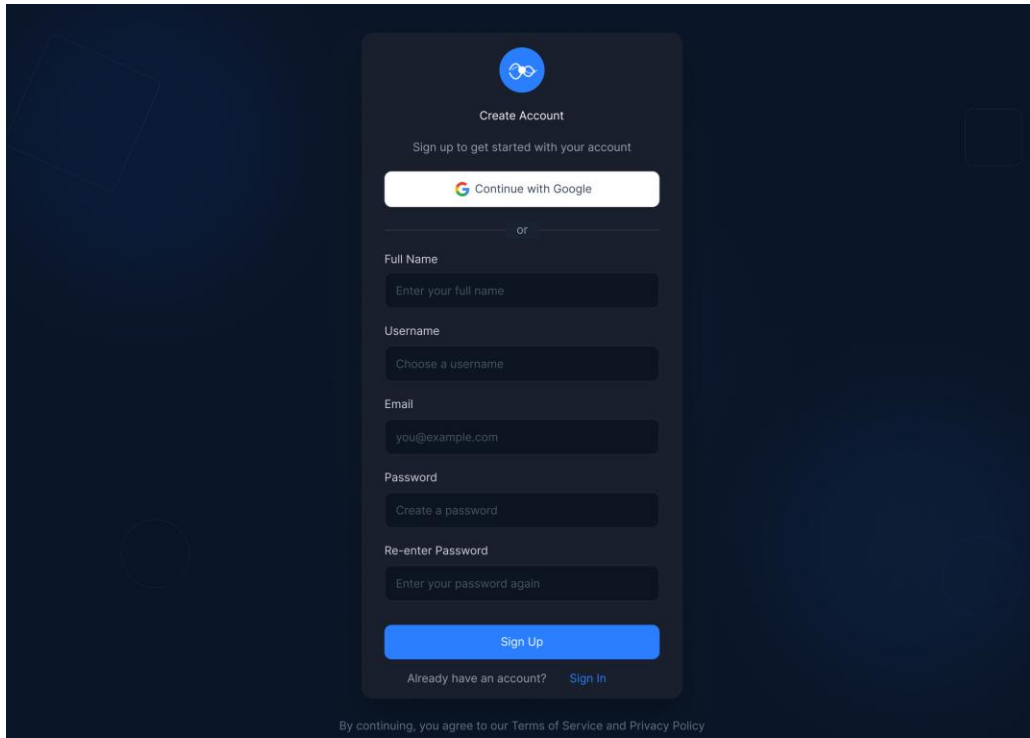
For the most detailed version of our prototype/mockup, please access this Figma link: <https://www.figma.com/design/rsUH7QorJeK7ahNM3BQ7T9/Mockup?node-id=0-1&t=7EirBBi6IRc1KLFC-1>

5.1 Authentication

5.1.1. Sign In Page

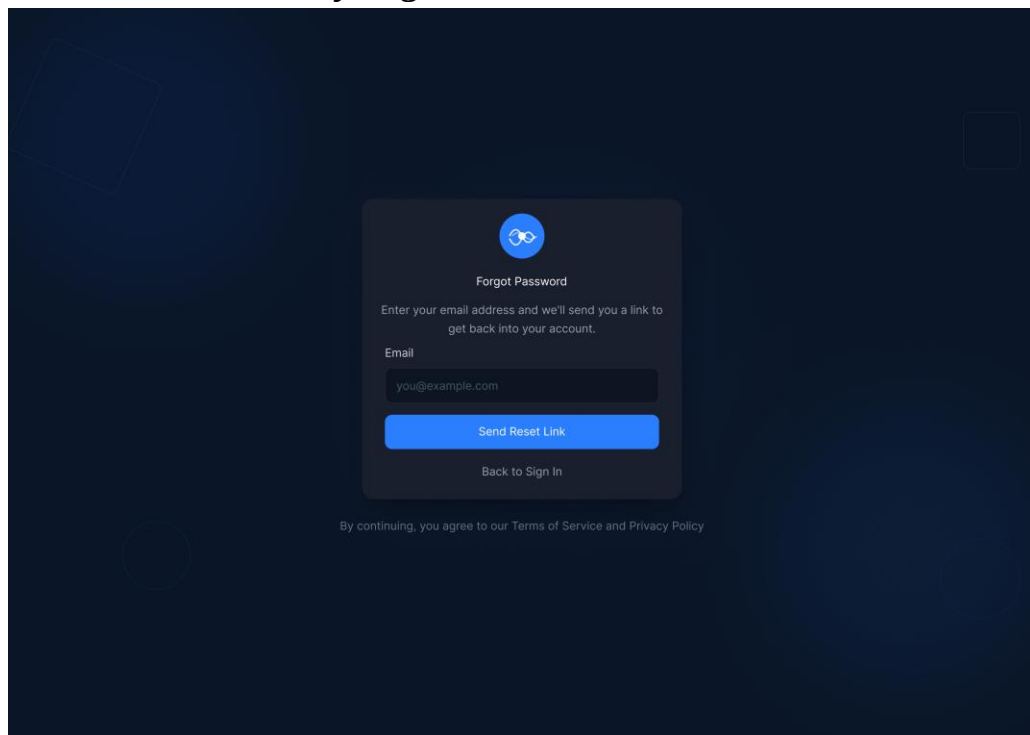


5.1.2. Sign Up Page



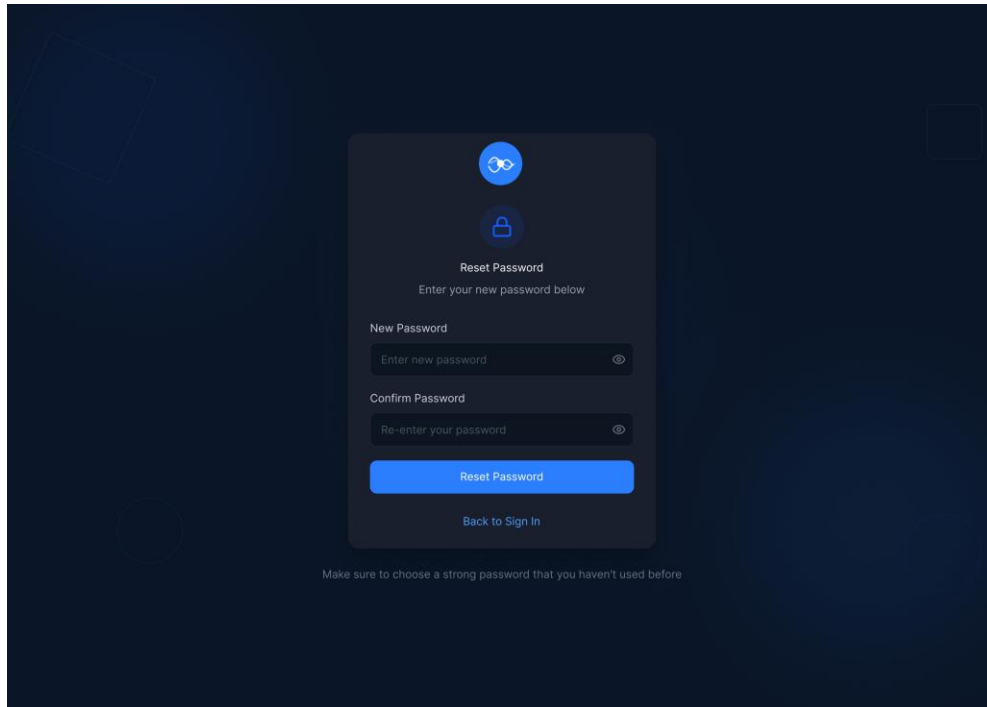
The Sign Up page features a dark blue background with a central white card. At the top of the card is a blue circular logo with a white '@' symbol. Below the logo, the text 'Create Account' is displayed, followed by the instruction 'Sign up to get started with your account'. A white button with the Google logo and the text 'Continue with Google' is positioned below the instruction. Underneath this button is the word 'or'. The form then includes several input fields: 'Full Name' with the placeholder 'Enter your full name', 'Username' with the placeholder 'Choose a username', 'Email' with the placeholder 'you@example.com', 'Password' with the placeholder 'Create a password', and 'Re-enter Password' with the placeholder 'Enter your password again'. A large blue 'Sign Up' button is located at the bottom of the form. Below the button, there is a link that says 'Already have an account? Sign In'. At the very bottom of the page, a small line of text reads 'By continuing, you agree to our Terms of Service and Privacy Policy'.

5.1.3. Password Recovery Page



The Password Recovery page has a dark blue background with a central white card. At the top of the card is a blue circular logo with a white '@' symbol. Below the logo, the text 'Forgot Password' is displayed, followed by the instruction 'Enter your email address and we'll send you a link to get back into your account.' Below this instruction is an 'Email' input field with the placeholder 'you@example.com'. A large blue 'Send Reset Link' button is positioned below the input field. At the bottom of the card, there is a link that says 'Back to Sign In'. At the very bottom of the page, a small line of text reads 'By continuing, you agree to our Terms of Service and Privacy Policy'.

5.1.4. Reset Password Page

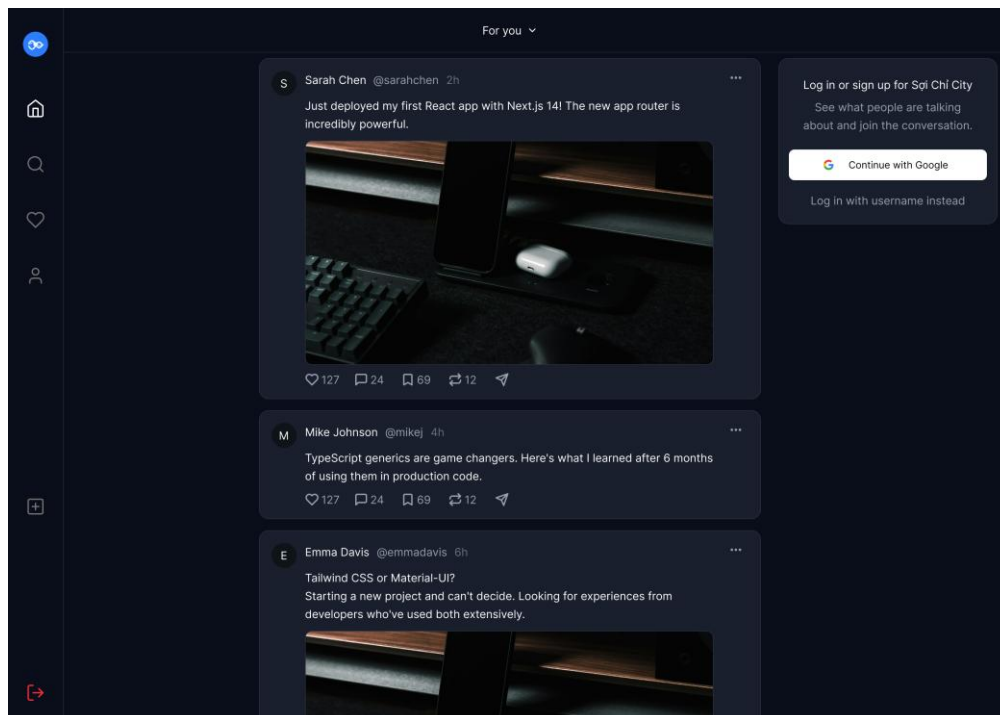


The image shows a 'Reset Password' form on a dark blue background. The form is centered and contains the following elements:

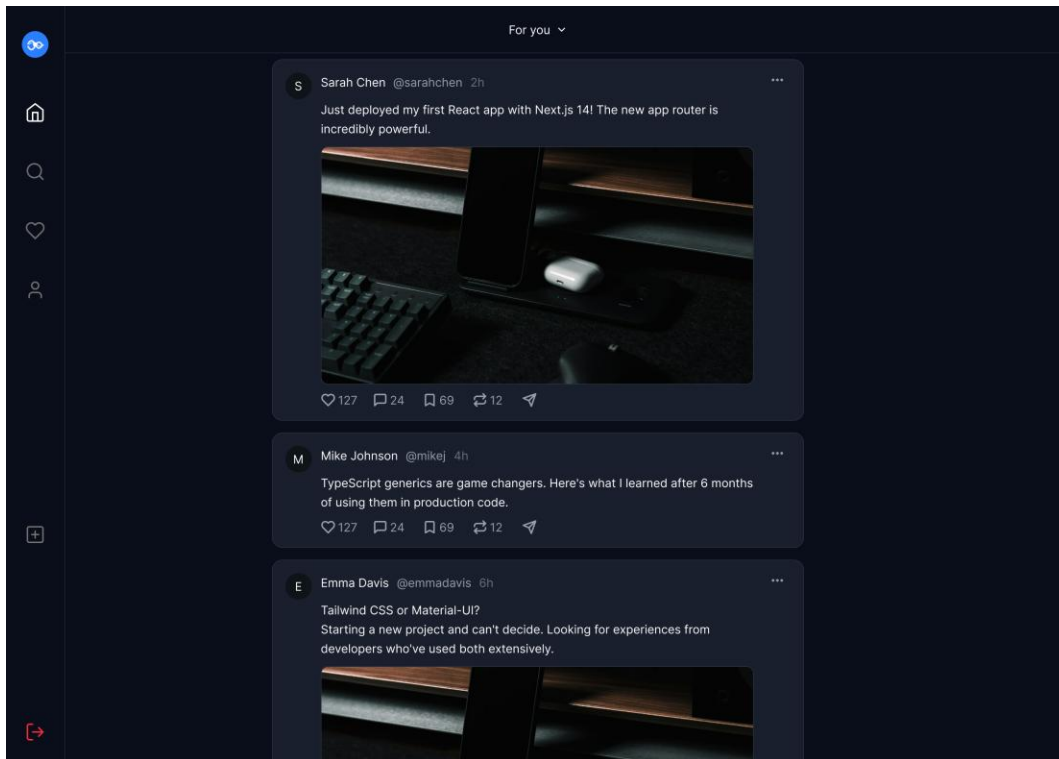
- A blue circular icon with a white keyhole symbol.
- A blue circular icon with a white padlock symbol.
- The text 'Reset Password' in white.
- The text 'Enter your new password below' in white.
- A section titled 'New Password' with a text input field containing the placeholder 'Enter new password' and a blue eye icon for toggling visibility.
- A section titled 'Confirm Password' with a text input field containing the placeholder 'Re-enter your password' and a blue eye icon for toggling visibility.
- A blue button labeled 'Reset Password'.
- A link labeled 'Back to Sign In'.
- A footer note: 'Make sure to choose a strong password that you haven't used before'.

5.2 Main Page

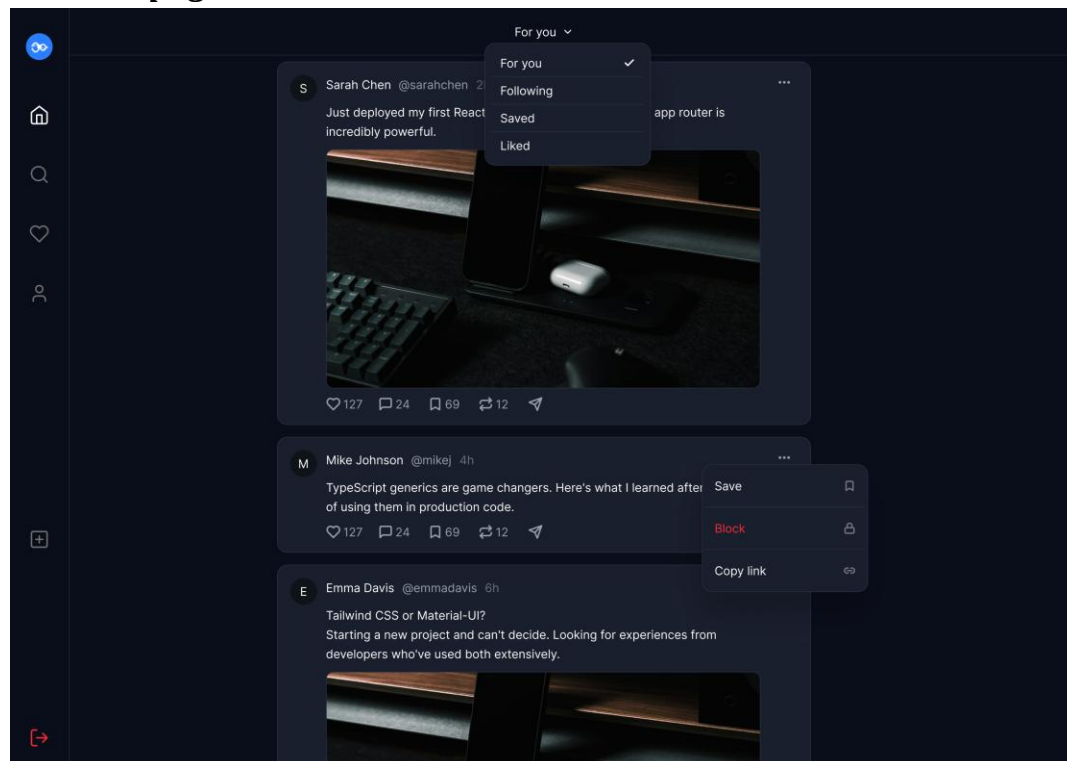
5.2.1. Feed page (guest)



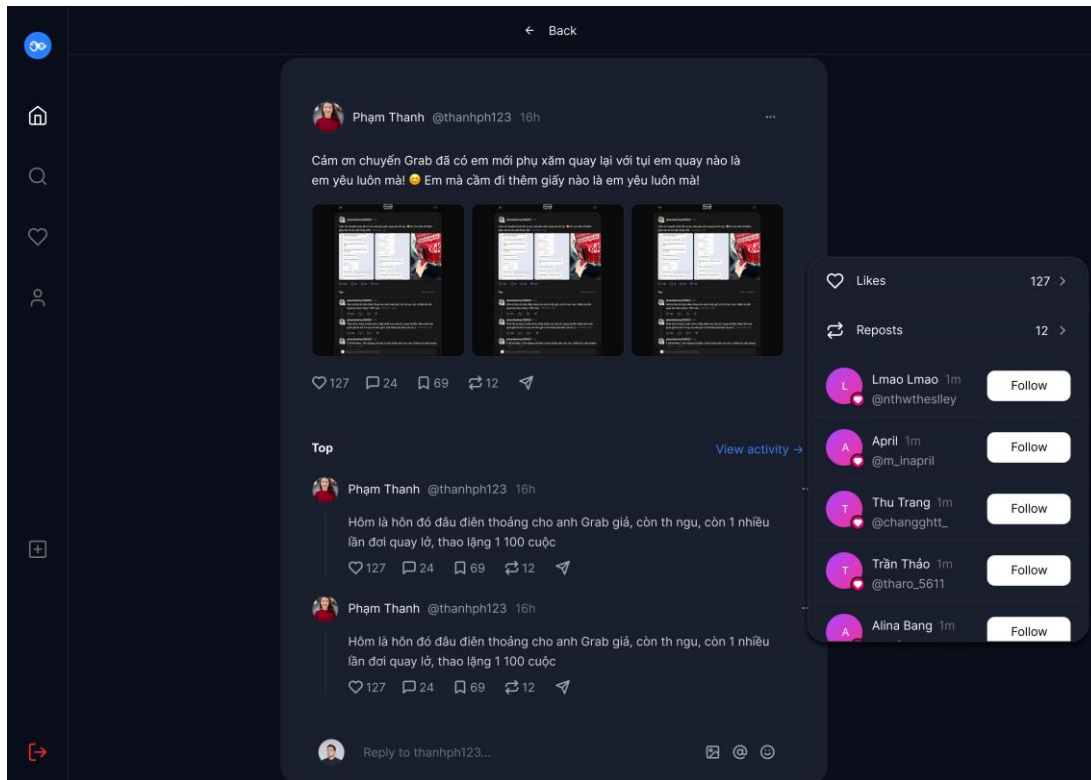
5.2.2. Feed page (registered user)



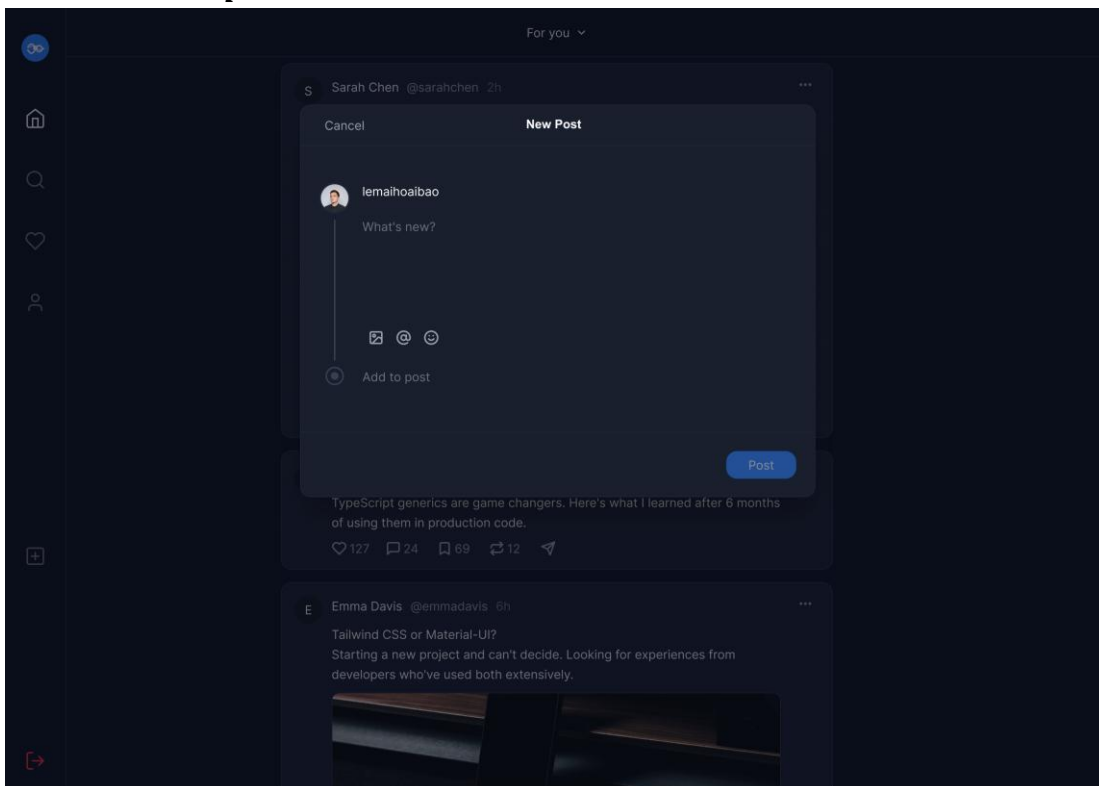
5.2.3. Feed page with interaction



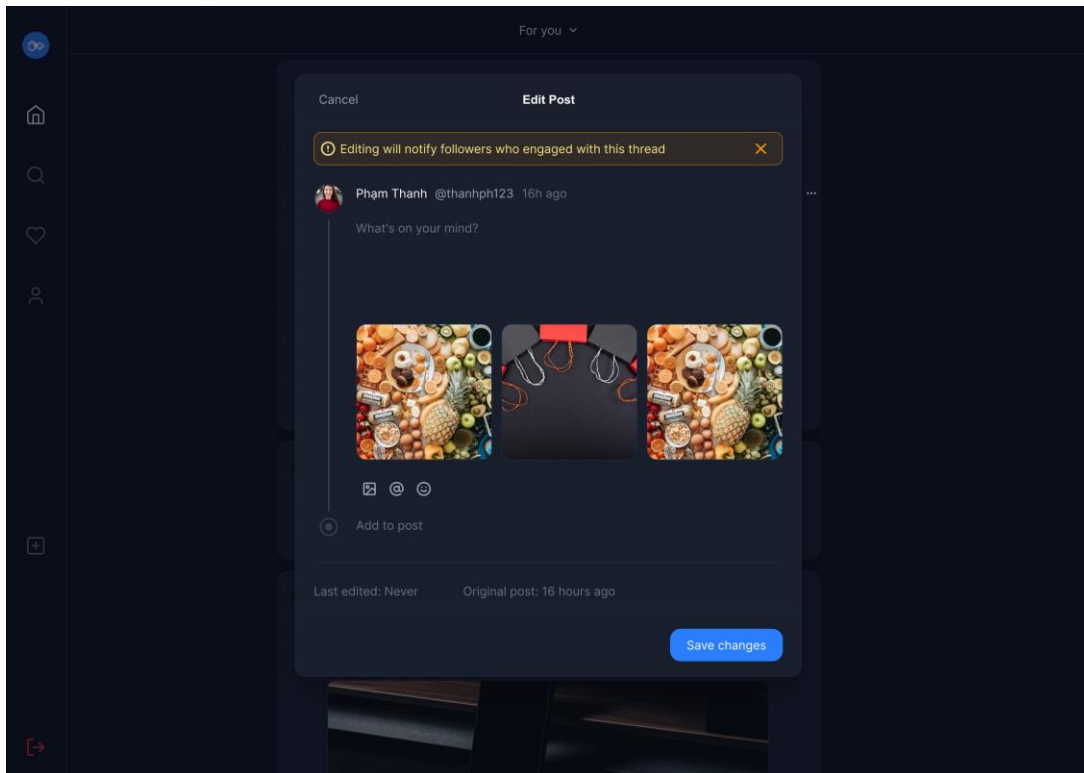
5.2.4. Detail post



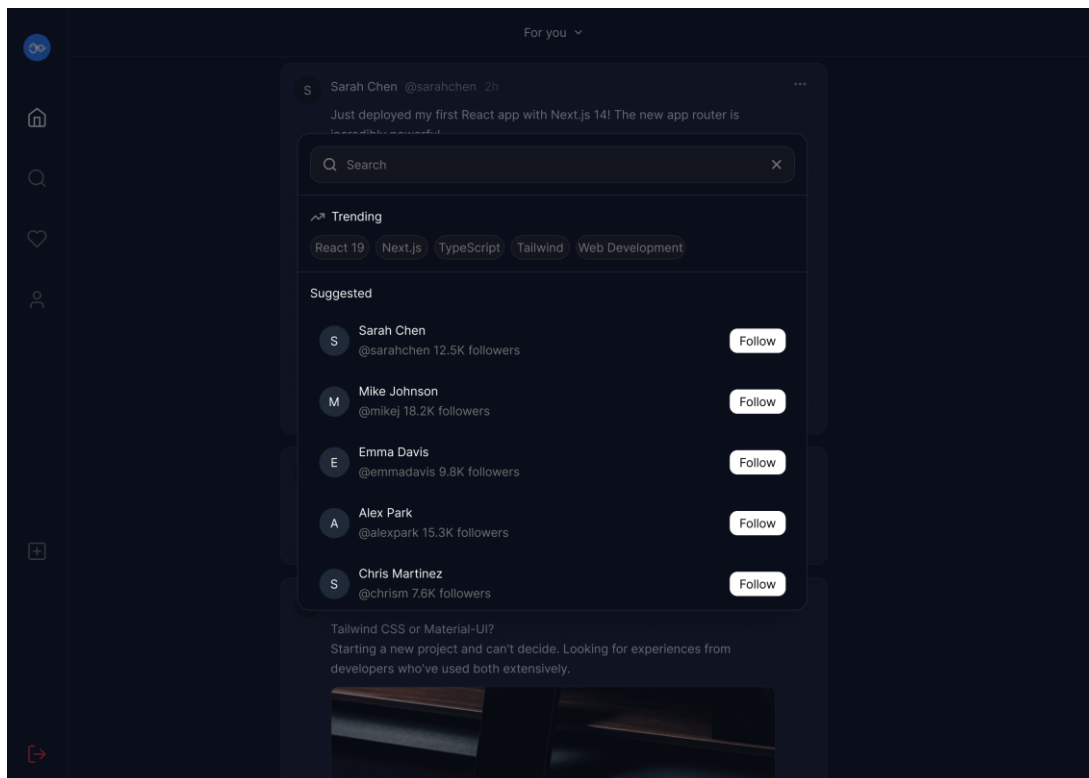
5.2.5. Create a post



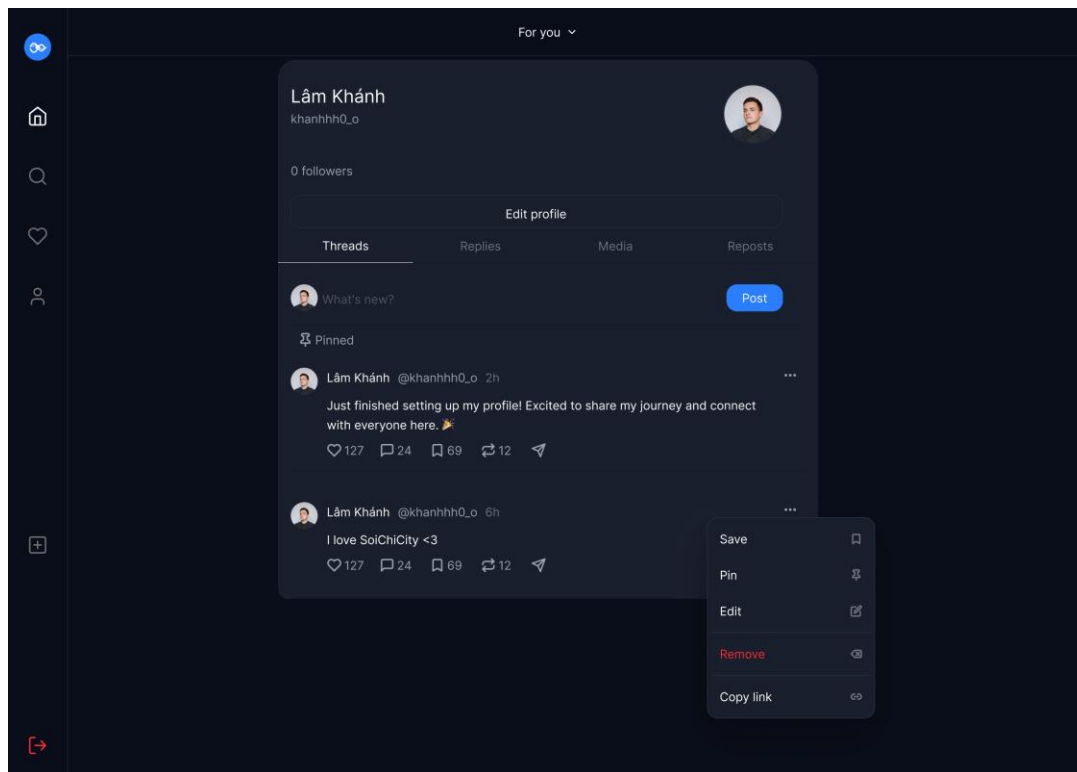
5.2.6. Edit my post



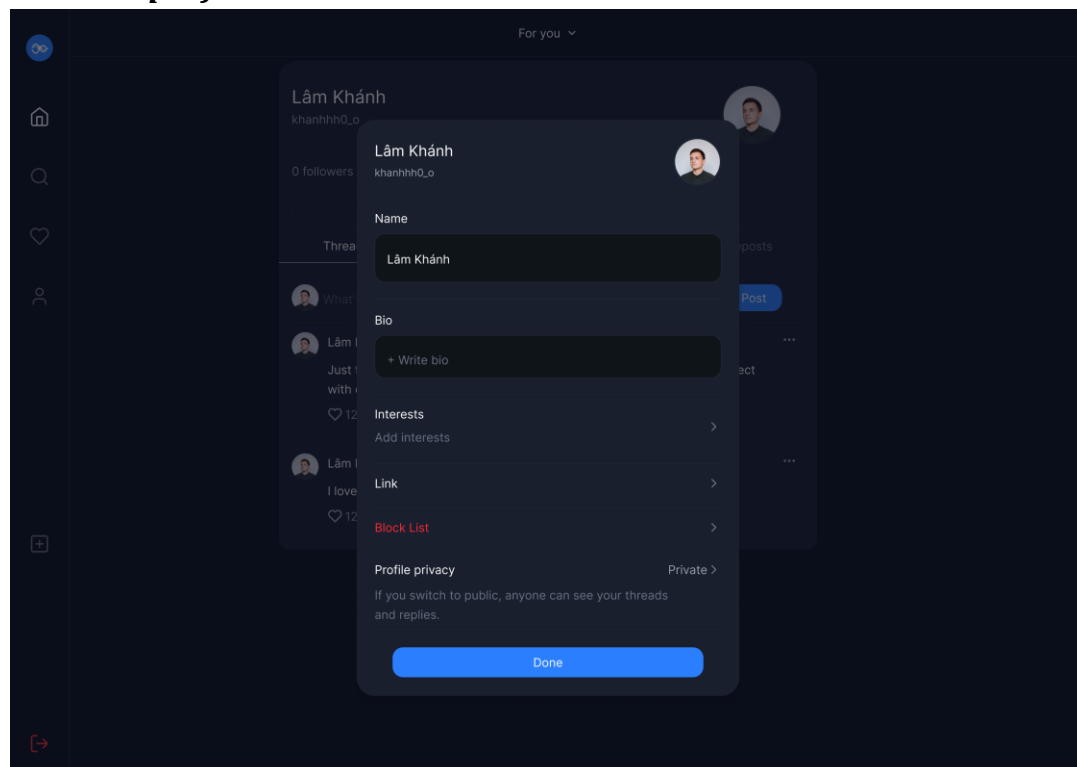
5.2.7. Search popup



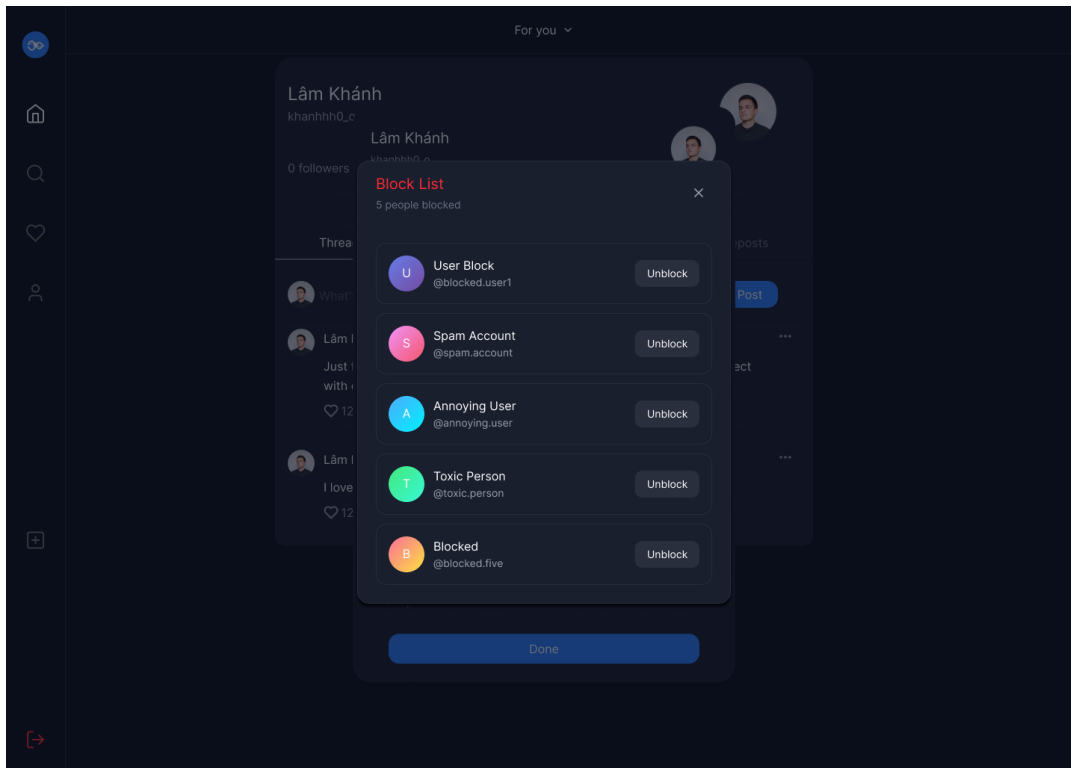
5.2.8. Owner Profile page



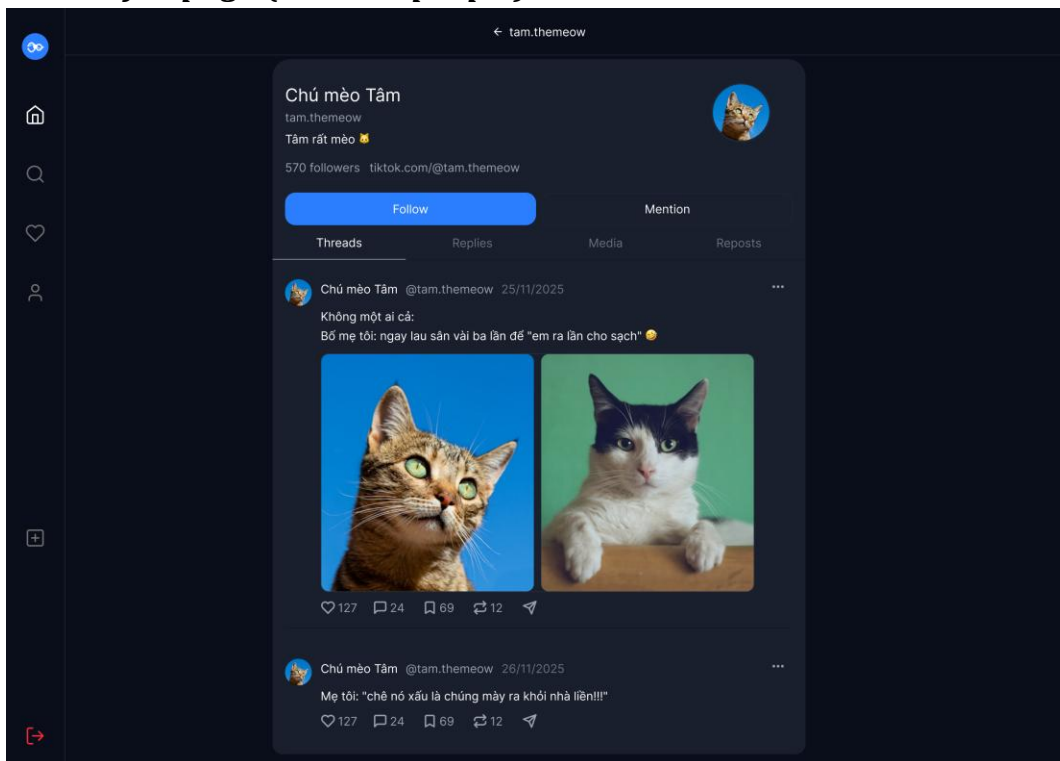
5.2.9. Edit profile



5.2.10. Block list popup



5.2.11. Profile page (another people)



5.2.12. Comment on post

